# RAG Architecture Notes

## 1. What is Retrieval-Augmented Generation?

RAG retrieves relevant documents and supplies them as context to a language model. The model answers from that context instead of relying only on what it memorised during training.

## 2. The indexing pipeline

The indexing pipeline runs once per document: Upload, Extract, Clean, Chunk, Embed, and Store in ChromaDB. Extraction turns files into text. Cleaning normalises whitespace. Chunking splits text into pieces of about 800 tokens with 120 tokens of overlap. Embedding converts every chunk into a numerical vector. The vectors, the chunk text and the source metadata are stored in a persistent ChromaDB collection.

## 3. Chunking versus embeddings

Chunking splits long text into smaller passages so that each passage fits the model's context window and focuses on one topic. Embeddings are different: an embedding is a list of numbers that represents the meaning of a chunk, so that chunks with similar meaning can be found by comparing vectors.

## 4. The question-answering pipeline

When a user asks a question, the same embedding model converts the question into a query vector. ChromaDB returns the top K chunks whose vectors are closest to the query vector. These chunks are placed in a prompt together with the question, and the chat model writes an answer with citations to the source pages.

## 5. Why overlap matters

Overlap repeats a small amount of text at the end of one chunk at the start of the next, so a sentence that falls on a boundary is still fully visible in at least one chunk.